

Quan Huynh

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EDUCATION

George Mason University

(Expected) May 2028

Bachelor of Computer Science GPA: 3.94/4.0

Coursework: Data Structures and Algorithms, Object-Oriented Programming (OOP), Web Development, Databases, Computer Systems, Artificial Intelligence, Statistics and Probability

EXPERIENCE

AI Software Engineer Intern

May 2026 – Sep 2026

Deepiri — AI R&D Collective

- Developed **Renderflow Studio**, an AI-powered video editing application, using **Tauri**, **Rust**, **TypeScript/Vite**, and **FastAPI**, building features for AI generation, timeline editing, and video processing
- Implemented AI job orchestration in **TypeScript** by binding tasks to projects, wiring **Accept/Reject** actions to **REST APIs**, and replacing polling with **Server-Sent Events (SSE)**, reducing job refresh latency by **60%**
- Developed video preview and active-clip scrubbing by integrating a custom **Rust** timeline engine with **FFmpeg**-based proxy frame extraction and native video streaming, reducing p95 preview startup latency to under **500ms**
- Built and evaluated AI workflows using models including SDXL-Turbo, Depth Anything V2, Qwen2.5-3B, and RIFE, optimizing local inference pipelines for Apple Silicon and cloud fallback environments.

University Learning Assistant

Aug 2025 – Dec 2025

George Mason University

- Assisted students during CS 112 Python labs by explaining programming concepts, debugging code, and guiding students through lab exercises.

PROJECTS

PaceTrack | *React.js, Node.js, Express.js, MongoDB, Socket.io, AWS S3, Mapbox GL JS*

May 2025 – Present

- Built full-stack running tracker with **React.js**, **TypeScript**, **Node.js**, and **Express.js**, supporting authentication, activity logging, club membership, and ownership-scoped authorization across **3 core data models**
- Implemented real-time club leaderboard system using **Socket.io** rooms, broadcasting saved runs to connected members and keeping rankings synchronized across concurrent client sessions with low-latency updates
- Designed GPS route ingestion pipeline with **AWS S3** presigned uploads, custom **GPX** parsing, haversine distance calculation, and elevation gain computation for server-side run analytics and route summaries

TradingLab | *React.js, TypeScript, NestJS, MongoDB, Redis, OpenAI API, Docker*

Apr 2026 – Present

- Built backend price service and portfolio backtesting APIs in **NestJS** and **TypeScript**, fetching historical market data from **CoinGecko** and **AlphaVantage** to support BTC, ETH, AAPL, and TSLA simulations
- Implemented multi-asset DCA portfolio workflow with **MongoDB**, weight-allocation validation, and reusable React asset components, enabling users to split recurring investments across assets with weights summing to **100%**
- Integrated AI advisor infrastructure using **OpenAI API**, **Server-Sent Events (SSE)**, and **Redis**-backed rate limiting, streaming real-time investment insights from portfolio diagnostics and backtest context

OpenCart | *Next.js, TypeScript, PostgreSQL, Prisma, Redis, Stripe Connect, NextAuth, AWS S3* Jun 2026 – Present

- Built full-stack multi-vendor marketplace with **Next.js**, **PostgreSQL**, and **Prisma**, supporting separate buyer and seller experiences across product listings, checkout, order tracking, and messaging
- Integrated **Stripe Connect** to onboard sellers and automatically split payments via destination charges, computing a **3%** platform fee per transaction and reconciling payment/order state through **Stripe webhooks**
- Developed real-time buyer-seller chat using polling-based message sync, product review system, and **AWS S3** image uploads for product listings, all secured with **NextAuth** session-based authorization

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, Go, C, Bash, HTML/CSS, SQL

Frameworks/Tools: Spring Boot, Flask, FastAPI, .NET, React.js, Next.js, Node.js, Express, Tailwind, Redux, gRPC

Databases: PostgreSQL, MongoDB, Redis, MySQL, DynamoDB, Supabase

Cloud/Infra: AWS (ECS, S3, RDS, Lambda), GCP, Docker, Kubernetes, Nginx, GitHub Actions, Terraform, Linux

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, LangChain, OpenAI API